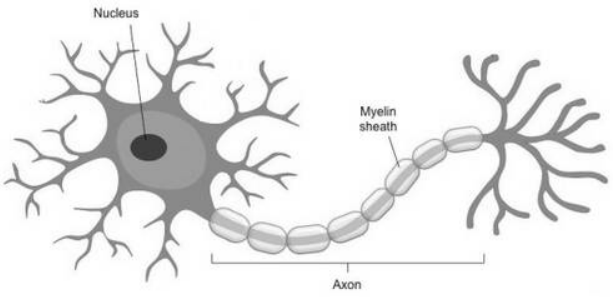
Type MS · Draw, label and compare the structure of a neurone

Answer 1 of 5 · Question 3 part a.i · 4 marks

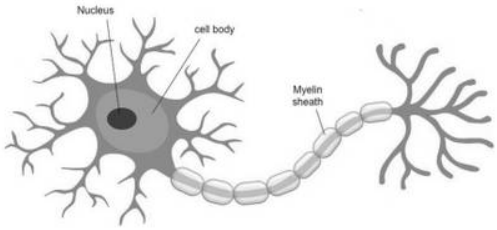
2025 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

3	(a)	(i)		1 mark for diagram 3 marks for labels	4
		(ii)	• transmits electrical/nerve impulses; • from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ;		2
	(b)	(i)	Any six from 1 (student X) holds up the ruler between the thumb and forefinger (of other student); 2 then drops/releases the ruler; 3 (other) student catches the ruler; 4 record the reading on the ruler/distance fallen when caught; 5 repeat for <u>this</u> student; 6 using the same distance between thumb and forefinger; 7 find mean/average value of reading on ruler; 8 calculate reaction time (using formula); 9 repeat with <u>other</u> students;		6
		(ii)	the reading on the ruler/ reaction time;		1
		(iii)	independent variable not continuous/is discrete;		1
		(iv)	reaction time = √(0.15 ÷ 5); reaction time = 0.17 (s);	accept 0.173(2)	2
				Total for Question 3 = 16 marks	

Type MS · Draw, label and compare the structure of a neurone

Answer 2 of 5 · Question 7 part b.ii · 4 marks

2024 JUN · Paper 01 (regional R) · find part b.ii in this mark scheme extract

7 (a) (i)	receptor (cell)		(1)
(ii)	sensory neurone		(1)
(b) (i)	a cell that has particular features/adaptations/structures; to carry out a certain/specific function in the body;		(2)
(ii)	diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending; nucleus; cell body; myelin sheath;  	Label marks can be awarded if they draw a sensory or relay neurone	(4)
(iii)	transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone; to effecter organs/muscle/gland;		(2)
(c)	100/4.0 x 10^{6}; = 0.000025 (s); = 0.025 (ms); = 2.5 x 10^{-2};	ecf for mps 2,3 and 4 Allow full marks for correct final answer	(4)

Total for Question 7 = 14 marks

Type MS · Draw, label and compare the structure of a neurone

Answer 3 of 5 · Question 1 part a.ii · 1 mark

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks

Type MS · Draw, label and compare the structure of a neurone

Answer 4 of 5 · Question 5 part a.ii · 1 mark

2020 JAN · Paper 01 · find part a.ii in this mark scheme extract

Number				
5	(a) (i)	12 x 50 = 600 (1) 600 ÷ 1000 = 0.6 (1)	Allow 600 without working out. Allow full marks for correct final answer.	2
	(ii)	provides insulation/prevents nerve impulse dispersing/speeds up nerve impulse (1)		1
	(b) (i)	• hormones are transported in blood (1) • speed of blood flow is determined by heart rate/hormones can only travel at the speed blood is moving/blood flow slower (1)	Allow reverse argument for both marking points	2
	(ii)	• recognition of distance divided by time (1) • $1.0 \div 1.2 \times 10^{2}$ / $1.0 \times 10^{6} \div 1.2 \times 10^{8}$ (1) • 0.0083(3) / 8.3×10^{-3} (s) (1)	Allow full marks for correct final answer	3

Total for Question 5 = 8 marks

Type MS · Draw, label and compare the structure of a neurone

Answer 5 of 5 · Question 5 part c.i · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

Number			
5 (a)	any three from • rapid response (to a stimulus) (1) • involuntary/automatic (1) • no brain involvement (1) • protective (1)		3 max
(b)	• A = motor (1) • B = relay/association/connector (1) • C = sensory (1)		3
(c) (i)	• fewer branches/dendrons/dendrites (1) • shorter branches/ dendrons/dendrites (1) • fewer knobs/synapses (1)	R axons	3
(ii)	• transmits impulse (1) • across synapse/(synaptic) cleft (1) • stimulates impulse in next neurone (1)		3
(iii)	• secreted by synapses/synaptic knob (1) • fewer synapses/knobs (1) • less secreted by remaining synapses/knobs (1)		3

Total 15 marks

Type MS · Draw, label and compare the structure of a neurone

Answer 5 of 5 · Question 5 part c.i · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks